



UNIVERSAL ORCHESTRATION

UX DESIGN FRAMEWORK

Key design considerations for Universal Orchestration Platforms

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Key design considerations for platforms where humans and digital workers share tasks, workflows, and accountability

1: Trust Gradient

2: Intervention Model

3: Shared Workspace

4: Info Hierarchy

5: Control Modes

6: Alert Design

7: Agency & Emotion

8: Cross-Device

THE FUNDAMENTAL DESIGN TENSION

Orchestration UI must serve **two radically different** cognitive **modes simultaneously**

The Worker

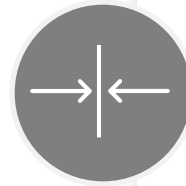
"What do I do next?"

Scanning their task queue for the next action

Wants clarity on priority and what's assigned to them

Needs to know what a digital worker has completed on their behalf

Low tolerance for noise — just tell me what matters to me



The Orchestrator

"What's happening — and where might it go wrong?"

Monitoring digital workers running in parallel

Looking for bottlenecks, failures, and escalation triggers

Needs visibility across both human and machine workstreams

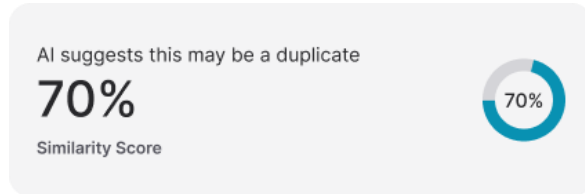
Needs to intervene with full context — fast

Key decision: Are these the **same person** or **different people**? This determines whether you build one surface or two.

Most platforms collapse them — and fail both users.

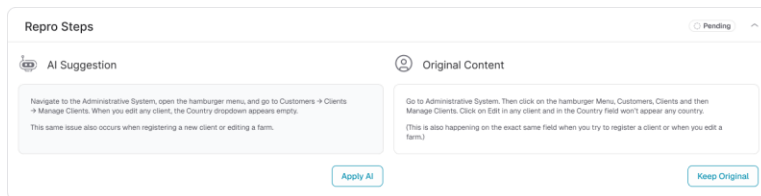
1: THE TRUST GRADIENT — MAKING AI WORK LEGIBLE

Users need to know not just THAT **a task was done**, but HOW **confident they should be in the result**



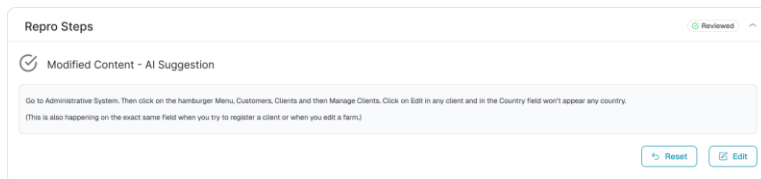
Confidence signaling at a glance

Every digital worker action needs a visible certainty indicator — surfaced inline, not buried in logs. A 95% confidence result looks and feels different from a 62% one. Color, weight, and iconography must differentiate these without the user having to dig.



Explainability on demand, not by default

Don't force AI reasoning onto every completed task — that recreates the doom loop. Offer a single tap to expose rationale. Power users want depth; casual users shouldn't be overwhelmed. Progressive disclosure is the pattern.



Distinguish action types by risk level

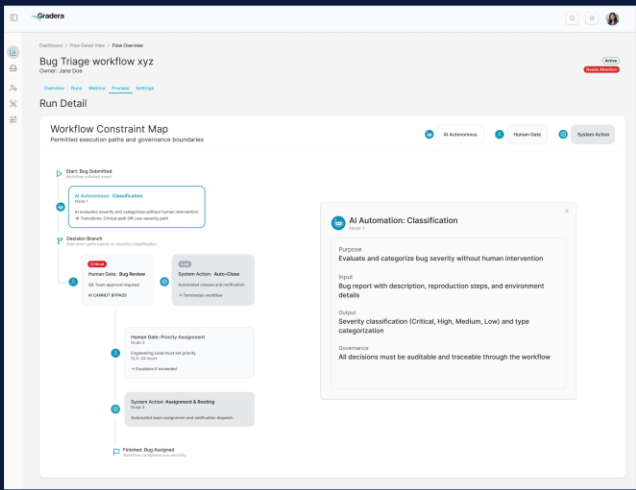
'Retrieved information', 'transformed data', 'sent externally', and 'made a decision' carry fundamentally different risk and reversal cost. The UI must surface this taxonomy — not just 'Task completed.' Risk tier should be visually immediate.

2: THE INTERVENTION MODEL — WHEN AND HOW HUMANS TAKE OVER

The **handoff** moment is the **hardest UX problem** in orchestration — and **most platforms handle it poorly**

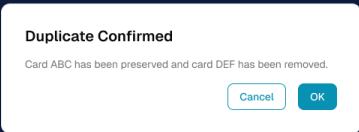
Warm handoffs over cold alerts

When a digital worker escalates to a human, it must hand off full context — what it tried, what it couldn't determine, what the human needs to decide, and what the downstream consequences of each option are. The human arrives 80% briefed, not cold.



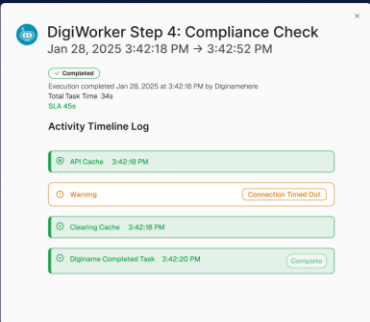
Decision gravity — friction proportional to consequence

Reversible decisions (approve a draft) and irreversible ones (send to 10,000 customers) should feel different. Apply confirmation states, previews, and cooling periods for high-stakes actions. Friction is not a UX failure — it's trust infrastructure.



Graceful degradation — the floor is always there

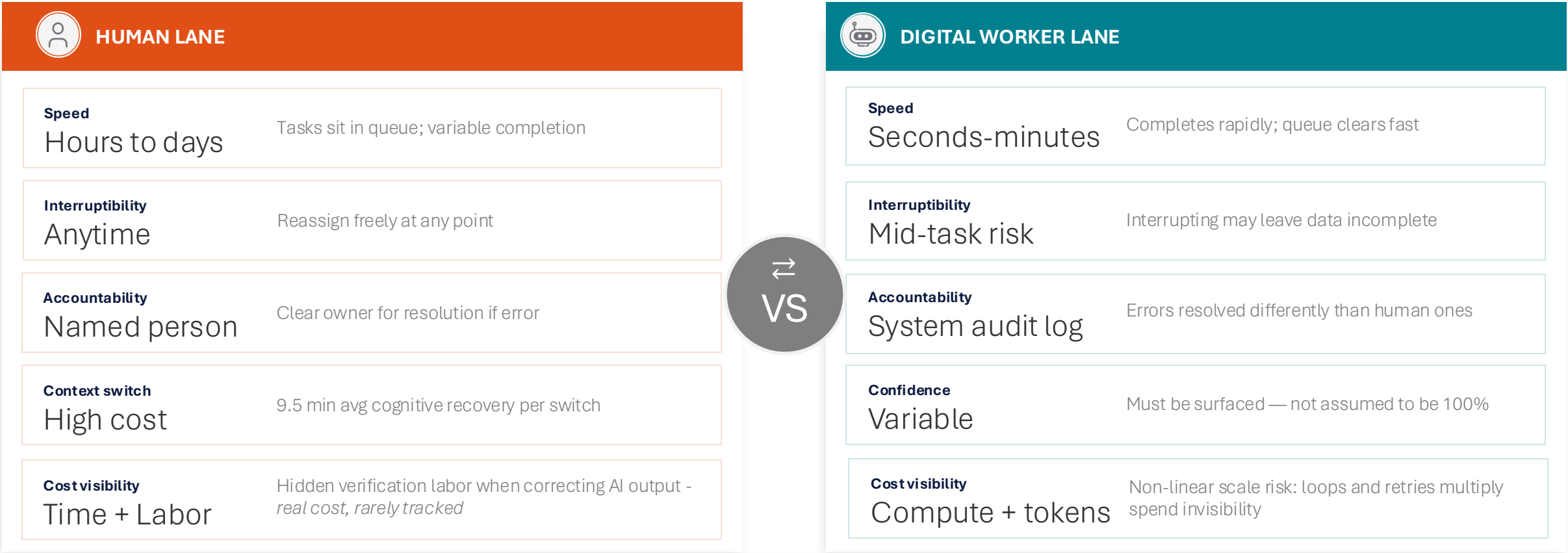
When a digital worker fails or gets stuck, the task must not disappear into an error state. It degrades gracefully into a human-assignable task with all context preserved. Users need to feel there is always a safety net beneath the automation.



3: THE SHARED WORKSPACE — HUMAN AND DIGITAL LABOR SIDE BY SIDE

The temptation is to make **human** and **digital tasks look identical**. *That is a mistake.*

THE TWO-LANE MODEL



Invisible meter: platforms show task done — not what it cost.
No signal to route tasks differently or cap digital worker spend.

Loop tax: agent routing loops bill per retry. Without visible timeout UX, a design gap silently becomes a spending incident.

Verification labor: human time correcting AI output is real cost — rarely tracked or attributed back to the digital worker.

Think less like a single Kanban board, more like an air traffic control display — managing flows with different physics

4: PROGRESSIVE DISCLOSURE & INFORMATION HIERARCHY

'Dashboard of everything' = most common orchestration failure. Structure info in 3 tiers — never mix into one list.

01

TIER 1 — ATTENTION REQUIRED NOW

Escalations · Blocked tasks · Human decisions pending

Maximum 5–7 items. If everything is urgent, nothing is. This list must be short, actionable, and never include FYI-level information. Users scan this first and leave if it's noisy.

Example

Digital worker blocked on ambiguous field — needs human decision to proceed

02

TIER 2 — IN MOTION

Active tasks · Progress status · ETA signals

What's running — both human and digital. Visible but not demanding attention. Users reference this tier to understand workload and pace. Status updates arrive here passively.

Example

3 digital workers processing invoices · 2 human tasks in review · 1 pending approval

03

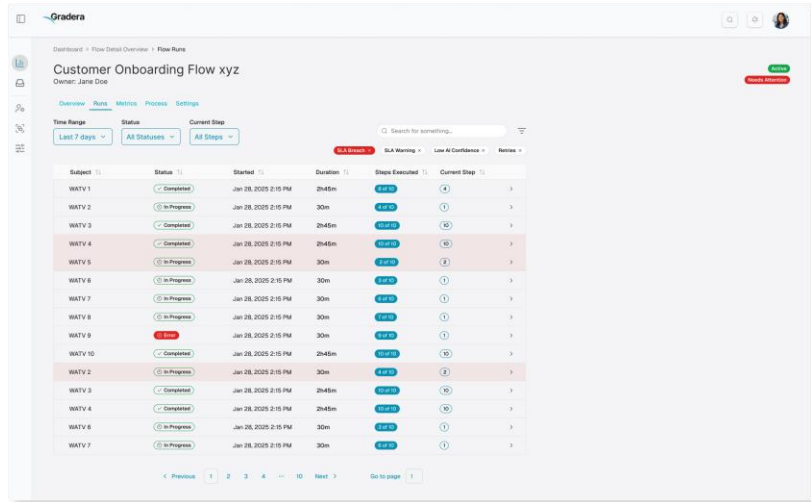
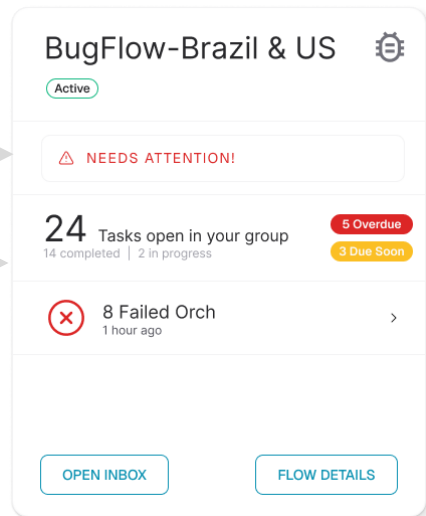
TIER 3 — COMPLETED / ARCHIVE

Audit trail · Outcome quality · Completion metrics

Accessible on demand, never foregrounded. Auditors, analysts, and post-incident reviewers live here. This tier should feel like a library — searchable, filterable, complete.

Example

847 tasks completed this week · 98.2% digital worker accuracy · 4 human overrides



5: CONTROL MODALITIES — FROM OBSERVER TO OPERATOR

Different users need radically **different control surfaces**. One omniscient view serves no one well.

Observer Mode

Executives · Auditors · Compliance

Read-only, high-level view of system health and outcomes. No controls visible — no temptation to act. Designed for situational awareness, not intervention.

Capabilities:

Dashboard metrics

Trend views

Export/report

Operator Mode

Workflow managers · Team leads · On-call

Real-time intervention surface. Pause, resume, reassign, approve. Escalation handling with warm context. High information density — designed for people actively watching.

Capabilities:

Pause/resume workers

Triage escalations

Override decisions

Builder Mode

Workflow designers · Admins · Architects

Workflow logic, trigger conditions, digital worker configuration. Complex and powerful controls — dangerous if misused. Should require deliberate mode-switch, never surface by accident.

Capabilities:

Edit trigger logic

Configure workers

Set thresholds

The **controls** appropriate for a **Builder** are **anxiety-inducing** for a **frontline Operator**. **Mode separation is not a feature — it's safety design.**

6: NOTIFICATION & ALERT DESIGN — AVOIDING FATIGUE

If digital workers **fire alerts** on **every completed step**, users develop **alert fatigue** within days and start **ignoring everything**

✗ WHAT BREAKS TRUST

- ✗ Alert on every digital worker task completion
- ✗ Undifferentiated urgency — everything looks the same
- ✗ Requires opening a separate dashboard to act
- ✗ No feedback loop — dismissed alerts keep firing
- ✗ Real-time noise on low-stakes background tasks

✓ WHAT BUILDS TRUST

- ✓ Exception-based alerting only — silence is good news
- ✓ Urgency tiers: Act Now / FYI / Digest visually distinct
- ✓ In-context actions: approve/reject without leaving workflow
- ✓ Alert fatigue scoring — track dismiss rates, auto-tune
- ✓ Batched digests at natural rhythms (start of day / EOD)

Design principle: **Users** should hear from the system **when something needs them** — **not when everything is going fine.**

Silence is the most reassuring signal an orchestration platform can send.

7: AGENCY, EMOTION & ACCOUNTABILITY — THE HUMAN LAYER

Workers feel **disempowered** when **AI acts on their behalf** without visible accountability.

Language frames the relationship

The interface copy matters enormously.

"The AI completed your task" = displacement.

"Your digital worker completed the task you assigned" = leverage.

Same outcome — fundamentally different psychological experience.

Visible credit attribution

When a human-digital collaboration produces a good outcome, both contributions should be visible in the record. If a digital worker prepped a document and a human closed the deal — that story should exist in the audit trail.

Easy override — always

Every digital worker action needs a visible, low-friction override path. Not because users will use it often — but because knowing it exists creates the psychological safety to let digital workers act autonomously.

The override button is trust infrastructure.

9% of workers trust AI for complex decisions vs **61% of executives**:

this gap is an **emotional design problem, not a technical one**

— WalkMe Digital Adoption 2026

8: CROSS-DEVICE & AMBIENT ORCHESTRATION

Orchestration work doesn't happen only at a desktop. The platform needs a coherent model across surfaces.



Desktop / Workstation

Full orchestration surface — all three modes available

High-density information views appropriate here

Builder and Operator controls live here exclusively

The primary surface for active workflow management



Mobile Triage Only

Approve / reject / reassign — nothing more

Cognitive load of orchestration decisions dramatically reduced

Escalation handling optimized for one-thumb interaction

No builder or configuration controls on mobile — ever



Ambient (Slack / Teams / Email)

Meet users where they already are for time-sensitive handoffs

One-tap approve/reject gets faster response than a dashboard

Digital worker escalations delivered in-thread with context

Minimize friction — don't require opening a separate app

State persistence across surfaces is table stakes:

if a user starts reviewing an escalation on mobile and picks it up on desktop, they must not have to re-orient.

THE NORTH STAR FOR ORCHESTRATION UX

Users should always **feel they have MORE control** and **BETTER visibility** than they would managing purely human workflows — **not less**.

EVALUATE EVERY DESIGN DECISION AGAINST THREE QUESTIONS:

1: Legibility

Does this help the user understand what's happening?

2: Trust

Does this help the user trust what's running?

3: Control

Does this help the user intervene when it matters?

SUMMARY

Having a UX framework in place is critical to a Universal Orchestration experience. It brings governance and control to this rapidly accelerating space. This UX governance provides an experience to the user that drives adoption, trust and peace of mind.

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8: Cross-Device



Thank You!

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